NOTICE OF
CHANGE

MIL-STD-35-74(MI)
NOTICE 4
18 SEPTEMBER 1992

MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

ACTUATORS, EXPLOSIVE

TO ALL HOLDERS OF MIL-STD-35-74(MI):

1. THE FOLLOWING PAGES OF MIL-STD-35-74(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
3	18 SEPTEMBER 1992	3	29 OCTOBER 1976
17	18 SEPTEMBER 1992	17	17 MAY 1974
18	17 MAY 1974	(Reprinted without change)	
33	18 SEPTEMBER 1992	33	17 MAY 1974
34	17 MAY 1974	(Reprinted without change)	
35	17 MAY 1974	(Reprinted without change)	
36	18 SEPTEMBER 1992	36	17 MAY 1974
71	18 SEPTEMBER 1992	71	17 MAY 1974
72	18 SEPTEMBER 1992	72	17 MAY 1974
75	18 SEPTEMBER 1992	75	17 MAY 1974
76	18 SEPTEMBER 1992	76	17 MAY 1974

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS.

3. Holders of MIL-STD-35-74(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
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Review activities:
Army-AT, AV, CR, ER
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User activity
Army - AR

Preparing activity:
Army - MI

(Project EGDS-A429)

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SECTION 1.1

SPECIFICATION TITLE INDEX

Specification Title Code	Specification Title	Applicable Specification
252001	Actuator, Linear, Explosive	MIL-A-46556
252002	Actuator, Rotary, Explosive	MIL-A-14812 MIL-A-14813 MIL-A-14897 MIL-A-14898 MIL-A-50372 MIL-A-50373 MIL-A-60606 MIL-A-60611 MIL-A-60657 MIL-A-60658 MIL-A-60681 MIL-A-60682 MIL-A-60692 MIL-A-60693 MIL-A-60837 MIL-A-60838 MIL-A-60848 MIL-A-60849 MIL-A-60888 MIL-A-60889

Supersedes page 3 and 4 of 29 OCTOBER 1976

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These should be completed on the SRS using the codes shown above. The write-ins should be in the exact words desired, but limited to 189 characters (including spaces and punctuation marks) for each characteristic code. When an additional requirement is added in Codes 7001 through 7005, an additional test is usually required (Codes 7011 through 7015).

Computer generated documents on actuators will automatically have a requirement to package them in accordance with the specification shown with Code 1006. If there are additional packaging requirements, these should be added through the use of Codes 7006 through 7010.

If additional notes are required in Section 6, these should be stated using Characteristic Codes 7016 through 7020.

Part marking shall be in accordance with the document shown with Code 1006. Additional marking requirements may be specified on the SRS with appropriate codes between 6991 and 6995.

1.2.9 Applicable Documents

Any specification or other document entered as a characteristic value for any Characteristic Code must also be listed with a code from Appendix B, in its proper numerical sequence. Only the number is required. Specification and/or document number entries shall be void of blank spaces; i.e., a number, letter, or punctuation mark must occupy every space from the beginning to the end of the entry. For example, MIL-A-55339 must be entered on the SRS as: 2004MIL-A-55339#. It is not necessary to list titles.

In addition to the above, add the below-listed documents when the indicated codes are used:

Document (Subject)		Code (Test)
MIL-STD-105 (Sampling Procedures and Inspection Tables)	7033	(Group A Sampling Plan)
	7044	(Group B Inspection)
MIL-STD-130 (Identification Marking)	1886	(Nameplate)

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Document (Subject)		Code (Test)
MIL-STD-202 (Test Methods)	1372, 1373	(Insulation Resistance)
	1412 thru 1417	(Terminal Strength)
	2504	(Circuit Resistance)
	4857	(Radiographic Inspection)
	4862	(Dielectric Withstanding Voltage at Sea Level)
	4885	(Dielectric Withstanding Voltage at Reduced Barometric Pressure)
MIL-STD-331 (Environmental and Performance Tests, Fuses)	2544	(Forty-Foot Drop)
MIL-STD-810 (Environmental Test Methods)	2004, 2006	(Altitude)
	2034, 2036	(High Temperature)
	2053	(Low Temperature)
	2074	(Temperature Shock)
	2090, 2092	(Sunshine)
	2114	(Rain)
	2136, 2138, 2140	(Humidity)
	2142, 2144	(Humidity)
	2184	(Fungus)
	2204	(Salt Fog)
	2234	(Dust)
	2256, 2258, 2260	(Explosive Atmosphere)
	2284, 2286	(Leakage, Immersion)
	2314	(Acceleration)
	2336	(Vibration)
	2386, 2388	(Acoustical Noise)
	2390, 2392	(Acoustical Noise)
	2414	(Shock)
	2454	(Combined Temperature-Humidity-Altitude)
MS16624 (Basic External Retaining Ring)	1826	(Retaining Rings)
MS16625 (Basic Internal Retaining Ring)	1826	(Retaining Rings)
MS20995 (Safety or Lock Wire)	1842	(Locking of Parts)
MS33540 (Safety Wiring and Cotter Pinning)	1842	(Locking of Parts)

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b. Enter an "X" with Code 1804 to specify that the direction of gas flow shall be indicated by arrows.

c. Enter an "X" with Code 1806 to specify that gas ports shall be closed with suitable and effective shipping caps or plugs when the actuator is not installed.

d. Enter a document number with Code 1808 to specify that the shipping caps or plugs shall conform to the entered document number. (See 1.2.9)

1.2.57 Fluid Connection Marking

Fluid connection markings may be specified on the SRS by entering an "X" with Code 1810 to specify that all ports for tube connections shall be clearly and permanently marked to indicate the connection to be made.

One or more of the following may also be used:

a. Enter an "X" with Code 1812 to specify that the direction of flow shall be indicated.

b. Enter an "X" with Code 1814 to specify that the use of abbreviations should be avoided.

c. Enter an "X" with Code 1816 to specify that decalcomanias shall not be considered permanent markings.

1.2.58 Seals

Seal requirements may be specified on the SRS by entering an "X" with Code 1818 to specify that the actuator shall be provided with suitable seals to preclude leakage of toxic gases.

A requirement that the seal shall be adequate to withstand the specified differential pressure cycling may also be specified by writing in the pressure value (psi) with Code 1820.

1.2.59 Retaining Rings

Retaining ring requirements may be specified on the SRS by entering an "X" with Code 1822 to specify that retaining or snap rings shall not be used where loss or dislodgement could adversely affect the operation of the actuator.

One or both of the following may also be used:

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a. Enter an "X" with Code 1824 to specify that neither retaining or snap rings shall be used in locations where the buildup of clearances and manufacturing tolerances will allow propellant gas leakage or end play contributing toward failure of parts.

b. Enter an "X" with Code 1826 to specify that retaining rings shall be used in accordance with Standards-MS16624 and MS16625. (See 1.2.9)

1.2.60 Bosses

Boss requirements may be specified on the SRS by writing in the drawing number with Code 1830 to specify that all internally threaded bosses shall conform to the specified drawing. (See 1.2.9)

One or both of the following may also be used:

a. Write in the drawing number with Code 1832 to specify that externally threaded connections shall conform to the specified drawing.

b. Enter an "X" with Code 1834 to specify that bosses shall be deep enough or shall incorporate fitting stops to prevent damage to the internal mechanism or shall not cause flow restrictions when fittings are screwed into the bosses to excessive depths.

1.2.61 Locking of Parts

Requirements for locking of parts may be specified on the SRS by entering an "X" with Code 1836 to specify that all threaded parts shall be locked by safety wiring, self-locking nuts, cotter pins, or other approved methods.

One or more of the following may also be used:

a. Write in the specification number with Code 1838 to specify that the self-locking nuts shall conform to a particular specification. (See 1.2.9)

b. Enter an "X" with Code 1840 to specify that mounting screws shall be drilled for lockwire.

c. Enter an "X" with Code 1842 to specify that lockwire shall be installed in accordance with Standard MS33540 and shall conform to Standard MS20995. (See 1.2.9)

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d. Enter an "X" with Code 1844 to specify that the use of lockwashers or staking is not permitted.

1.2.62 Movable End Fittings

Movable end fittings may be specified on the SRS by entering an "X" with Code 1846 to specify that the actuator shall be equipped with movable end fittings.

One or more of the following may also be used:

a. Write in the value (degrees) with Code 1848 to specify that the end fittings shall be capable of (Code 1848) degrees of rotational adjustment.

b. Write in the value (inches) with Code 1850 to specify that the end fittings shall be capable of (Code 1850) inches of linear adjustment.

c. Enter an "X" with Code 1852 to specify that the end fittings shall be equipped with self-aligning bearings.

1.2.63 Mechanical Stops

Mechanical stops may be specified on the SRS by entering an "X" with Code 1854 to specify that nonjamming mechanical stops shall be provided to limit overtravel at the end of the rated stroke.

1.2.64 Electrical Connections

Electrical connection conformance to a specification may be specified on the SRS by writing in the specification number with Code 1856. (See 1.2.9)

1.2.65 Electrical Wiring

Electrical wiring requirements may be specified on the SRS by writing in the specification number with Code 1858 to specify the specification to which electrical wiring is to conform. (See 1.2.9)

One or both of the following may also be used:

a. Enter an "X" with Code 1860 to specify that all leads shall be firmly clamped and potted.

b. Write in the minimum wire size value (AWG) with Code 1862 to specify that wire less than (Code 1862) AWG shall not be used.

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1.2.66 Electrical Connectors

Electrical connector conformance to a specification may be specified on the SRS by writing in the specification number with Code 1864. (See 1.2.9)

1.2.67 Shear Pins

Shear pin requirements may be specified on the SRS by writing in both the low and the high temperature values (degrees C) with Codes 1866 and 1868, respectively, to specify that shear pins shall be of a material which demonstrates uniform shear strength characteristics over the specified temperature range.

1.2.68 Markings

Markings may be specified on the SRS by entering a document number with Code 1870 to specify that the actuator markings shall be made with marking ink conforming to the entered document. (See 1.2.9)

Data to be marked on the actuator may also be specified by selecting one or more of the following requirements and writing in the information with the appropriate code(s):

Data Required	Code
Item name	1872
Manufacturer's part no.	1874
MARK and MOD nomenclature	1876
Date of manufacture	1878
Manufacturer's symbol	1880
Loading date	1882
Lot number	1884

1.2.69 Nameplate

Nameplate requirements may be specified on the SRS either by entering an "X" with Code 1886, to specify that each actuator shall be clearly and permanently identified by a durable nameplate conforming to MIL-STD-130 (see 1.2.9), or by writing in the drawing number with

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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

CHARACTERISTIC

REF. PARA.	NAME	CODE
1.2.55 CONT.	SUBASSEMBLIES SHALL BE TIGHTENED SUFFICIENTLY TO PRECLUDE LOOSENING OR DISASSEMBLY BY HANDLING, VIBRATION, AND SHIPMENT	1775
	THREAD DIAMETER (INCHES) AND LARGER	1778
	*TORQUE (INCH-POUNDS)	1776
	THREAD DIAMETER (INCHES) AND SMALLER	1782
	*TORQUE (INCH-POUNDS)	1780
1.2.56	GAS PORTS	
	THE MATING OF INLET PORT SHALL BE DIFFERENT FROM OPPOSITE END	1800
	GAS PORTS TO BE LABELED "INLET" AND "OUTLET", AS APPLICABLE	1802
	DIRECTION OF GAS FLOW SHALL BE INDICATED BY ARROWS	1804
	GAS PORTS SHALL BE CLOSED WITH SUITABLE AND EFFECTIVE SHIPPING CAPS OR PLUGS WHEN ACTUATOR NOT INSTALLED	1806
	SHIPPING CAPS OR PLUGS CONFORM TO (DOC DES)	1808

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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

CHARACTERISTIC

REF. PARA.	NAME	CODE
1.2.57	FLUID CONNECTION MARKING	
	ALL PORTS SHALL BE CLEARLY AND PERMANENTLY MARKED TO INDICATE CONNECTION TO BE MADE	1810
	DIRECTION OF FLOW SHALL BE INDICATED	1812
	USE OF ABBREVIATIONS SHOULD BE AVOIDED	1814
	DECALCOMANIAS NOT CONSIDERED PERMANENT MARKINGS	1816
1.2.59	SEALS	
	ACTUATOR SHALL BE PROVIDED WITH SUITABLE SEALS TO PRECLUDE LEAKAGE OF TOXIC GASES	1818
	SEAL SHALL WITHSTAND DIFFERENTIAL PRESSURE CYCLING OF (PSI)	1820
1.2.59	RETAINING RINGS	
	SHALL NOT BE USED WHERE LOSS OR DISLODGEMENT COULD ADVERSELY AFFECT ACTUATOR OPERATION	1822
	NOT TO BE USED IN LOCATIONS WHERE BUILDUP OF CLEARANCES AND MANUFACTURING TOLERANCES WILL ALLOW PROPELLENT GAS LEAKAGE OR ENDPLAY CONTRIBUTING TOWARD FAILURE OF PARTS	1824
	USE IN ACCORDANCE WITH STANDARDS MS16624 AND MS16625	1826

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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

CHARACTERISTIC

REF. PARA.	NAME	CODE
1.2.67	SHEAR PINS	
	MATERIAL SHALL DEMONSTRATE UNIFORM SHEAR STRENGTH CHARACTERISTICS OVER TEMPERATURE RANGE OF	
	LOW TEMPERATURE (DEG C)	1866
	*HIGH TEMPERATURE (DEG C)	1868
1.2.68	MARKINGS	
	MARKINGS SHALL BE MADE WITH BLACK MARKING INK CONFORMING TO (DOC DES)	1870
	DATA TO BE MARKED ON ACTUATOR (SELECT AS REQUIRED)	
	ITEM NAME	1872
	MANUFACTURER'S PART NUMBER	1874
	MARK AND MOD NOMENCLATURE	1876
	DATE OF MANUFACTURE	1878
	MANUFACTURER'S SYMBOL	1880
	LOADING DATE	1882
	LOT NUMBER	1884

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SPECIFICATION REQUIREMENTS

SEE MIL-STD-35 FOR EXAMPLE OF VALUE FILL-IN.

CHARACTERISTIC

REF. PARA.	NAME	CODE
1.2.69	NAMEPLATE (SELECT ONE)	
	IDENTIFIED BY DURABLE NAMEPLATE CONFORMING TO MIL-STD-130	1886
	OR	
	IDENTIFIED BY A DURABLE NAMEPLATE CONFORMING TO DRAWING ____.	1888
	DATA TO BE INCLUDED ON NAMEPLATE (SELECT AS REQUIRED)	
	ITEM NAME	1890
	MANUFACTURER'S PART NUMBER	1892
	MARK AND MOD NOMENCLATURE	1894
	DATE OF MANUFACTURE	1896
	MANUFACTURER'S SYMBOL	1898
	LOADING DATE	1900
	LOT NUMBER	1902
1.2.70	DEMAGNETIZATION	
1.2.70.1	TEST PROCEDURE	
	TEST ACTUATOR FOR RESIDUAL MAGNETISM USING A STANDBY COMPASS AND THE TEST SETUP SHOWN IN FIGURE 252-03-1.	2530

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